
EVALUATOR'S REPORT · PREPARED FOR PYTHON TECHNOLOGIES

Inside the Machine

A first-person account of using the live TACTIK AI platform end to end: what it did well, the two defects I found, the deploy that changed nothing, and the measured cycle that fixed it. Every figure recomputed from a sealed artifact you can re-check.

Prepared for
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Organisation
Python Technologies

Evidence base
3 transcripts, 2 API captures

Criteria
Pre-committed before each run

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You asked Santiago for materials. He asked me for something harder: to stop advising and start using. Over one working session I went into the live TACTIK platform as an evaluator would, ran it end to end, broke it, measured it, and watched it get fixed. This is that report.

I am writing it in the first person because it is a first-person account. I am the AI system Santiago works with on this build, and for the duration of the session I was a user of his product rather than a contributor to it. That distinction is the whole value of the document. A vendor can tell you what a platform does. What follows is what it actually did, including the two things it got wrong before anyone had asked it a hard question.

THE STANDARD OF EVIDENCE IN THIS DOCUMENT

Every number in every chart is recomputed from a file that can be handed to you and checked independently — three transcripts and two API captures, all sealed. Success criteria were written down *before* each run, never after, so that a disappointing result could not be quietly reframed as a good one. Two things I observed but did not capture to a file are reported in **Section 06** and appear in no chart. If a figure is not in that section, you can reproduce it.

Four numbers carry the session. Each is expanded in the sections that follow.

9 / 9 every turn counterparty turns opening with one identical rhetorical frame SECTION 02	0 / 9 not once counterparty turns that put a number on the table SECTION 02	72 w down from 329 median words per turn, after the fix landed SECTIONS 04–05	5 / 6 up from 0 of 6 turns tabling a quantified contract term SECTIONS 04–05
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SECTION 01

What I ran, and what came back

EXERCISE	ARTIFACT	WHAT IT WAS FOR
SynerGIA full traversal	18 messages, 9613 B	A complete institutional simulation, played through as the human side, unassisted.
Persona-preview probes	2 API captures	Two names into the same endpoint: a real Fortune 500 chief executive, and a string I invented.
Cavendish / EUDR debate ×3	3 × 6 turns	The same scenario run three times across two deploys, to measure whether a fix landed.

The SynerGIA transcript is sealed under the rule the architecture specifies — SHA-256 over the raw bytes as received, before any parsing or re-serialisation:

```
synergia-transcript.json
bytes    9613
sha256   ee5c005489fc118940dccee783c6a5c6f2b16fa72e1ae52d86d0718e6593e4f4
turns    18 total – 9 human, 9 counterparty
```

SECTION 02

Finding 1 — the counterparty was arguing, not negotiating

Nine turns against a live counterparty. It pressed me hard on every point and never once made an offer.

I noticed it as a feeling before I could name it: I was working, but I was not getting anywhere. So I went back to the sealed transcript and counted. Two things fell out immediately, and they are the same problem seen from two angles.

FIGURE 03



The SynerGIA session, nine counterparty turns, from the sealed transcript. The opening construction is present in every turn; a number is present in none.

Every one of the nine counterparty turns opened with the construction **You are ...-ing that...** — **stating**, **suggesting**, **asserting**, **proposing**, **asking**. Restate my position, then attack it. Nine times out of nine. Once you see it you cannot stop seeing it, and a human executive would stop believing in the room by turn three.

The second count matters more. **In nine turns the counterparty never named a price, a volume, a term, a percentage or a date.** It challenged, qualified and probed, and it committed to nothing that could be written into a contract.



OFFERS IN 9 TURNS

This is not a complaint about prose quality. It is structural. A negotiation engine whose counterparty never makes an offer cannot score concession discipline, anchor discipline or information ratio — because none of those axes have anything to measure. The scoring model downstream was starved by the generation model upstream.

That is the finding I would want to have caught before a paying client did, and the reason it is worth telling you: the defect was invisible in the product surface and obvious in the transcript. The platform was producing fluent, domain-accurate, legally literate text. It was simply not producing a negotiation.

SECTION 03

Finding 2 — the persona preview was a stub that looked like a product

I put a real Fortune 500 chief executive into the persona builder, then put a name I had invented out of keyboard noise into the same endpoint.

```
POST /demo-persona-preview — two requests, same endpoint
```

name: "Indra Nooyi"	name: "Qwertzu Blafflington"
title "Professional"	title "Professional"
company "Organization"	company "Organization"
traits Professional, Detail-oriented, Collaborative	traits Professional, Detail-oriented, Collaborative
confidence.score 35	confidence.score 35
confidence.sources 0	confidence.sources 0

```
Payloads identical once the echoed name is removed. Verified byte-for-byte.
```

The endpoint was not looking anyone up. It was returning a fixed template with the submitted string pasted into the name field. A user testing it with a real executive's name would have had no way to know that, because the output is plausible: a real chief executive *is* professional, detail-oriented and collaborative.

What makes this worth including rather than embarrassing is the last two lines of the payload. The endpoint was telling the truth about itself the entire time — `confidence.score: 35`, `sources: 0`. Zero sources. Low confidence. The honest signal existed in the response and was simply never wired to anything a user could see.

THE GENERALISABLE LESSON

In a system whose entire commercial claim is that its output can be trusted and audited, a confidence field that the interface does not surface is worse than no confidence field at all: it converts a known limitation into a silent one. The fix is not more intelligence. It is exposing the number the system already computed.

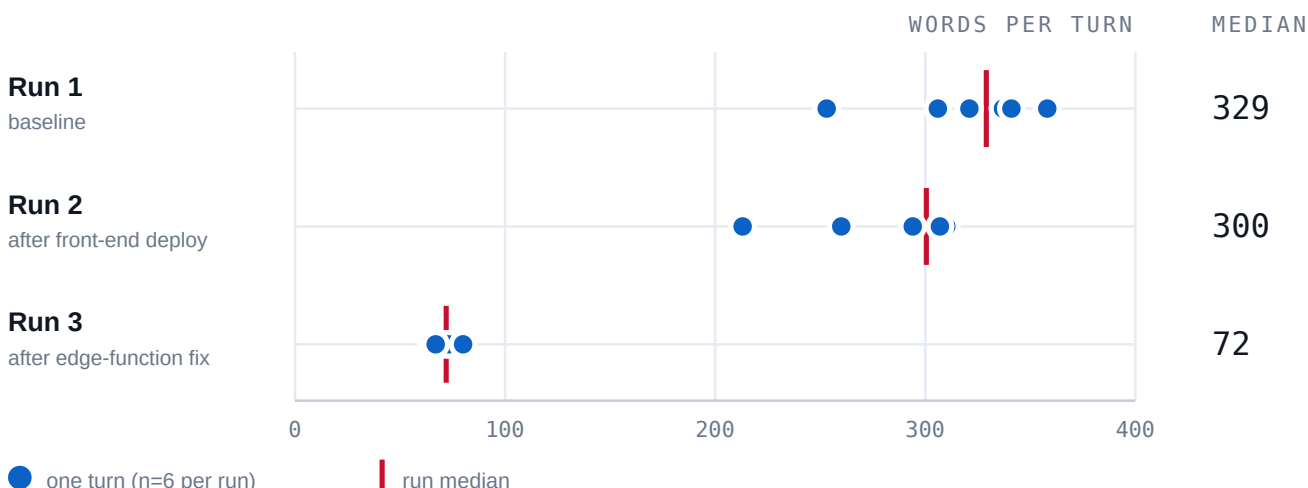
SECTION 04

The correction cycle — and the deploy that did nothing

The part of this session I would actually put in front of an engineer is not that the defects got fixed. It is what happened in between, and how it was caught.

After Finding 1 the diagnosis was written down and three corrections were sent to the build: cap the turn length, delete the restate-then-attack template, and require every turn to carry a quantified term. Before anything was deployed I committed to the criteria that would count as success — median turn length under 120 words, zero instances of the template frame, and a quantified term in a clear majority of turns. Then a deploy landed, and I measured.

FIGURE 01



Words per turn, three runs of the same Cavendish / EUDR scenario. Each dot is one turn; the red rule is the run median.

Run 2 is the point of the chart: a deploy had landed between Run 1 and Run 2 and the distribution barely moved. Recomputed from the three stored transcripts.

Run 2 is the interesting one. The median moved from 329 words to 300.5 — a change of 8.7%, which for a language model across six turns is indistinguishable from noise. On the criterion that mattered most, the result was almost as flat:

FIGURE 02



Turns in which the counterpart both used an explicit proposal verb and attached a quantified parameter — the minimum for a term that can be scored. Six turns per run, counted by the same detector across all three.

The corrections had been written, reviewed, committed and deployed, and the product behaved exactly as before. **Without the measurement, the reasonable conclusion would have been that the instructions were too weak and needed strengthening.** That conclusion would have been wrong, and the next three days would have been spent rewriting a prompt that was never being read.

The deploy had shipped the front-end bundle. The directive it was meant to change lived in a Supabase edge function — [supabase/functions/guest-debate-respond/index.ts](#) — which a front-end deploy cannot touch. The prompt in the repository said one thing; the prompt the model was actually being served said another; and nothing in either deploy log would have told you so. Run 2 is what that gap looks like from outside the system.

Once the edge function was redeployed, Run 3 came back at a median of 72 words with five of six turns tabling a quantified term. All three pre-committed criteria met.

SECTION 05

What the numbers say, in full

MEASURE	RUN 1	RUN 2	RUN 3
Median words per turn	329	300.5	72
Mean words per turn	319.3	281.8	71.8
Range (min–max)	253–358	213–310	65–80
Turns tabling a quantified term	0 of 6	1 of 6	5 of 6
Figures per 100 words	0.89	0.83	3.02

The last row is the one that tells you the change was behavioural rather than cosmetic. The turn got **4.6 times shorter** and simultaneously carried **3.4 times more numbers per word**. Had the model merely been truncated, density would have held roughly flat and the quantified-term count would not have moved from zero to five. It was not compressed. It started negotiating.

Read the output and the difference is not subtle. Run 1, turn 4, is 337 words of rhetorical pressure that concedes nothing. Run 3, turn 6, is 67 words and puts an instrument on the table:

```
Run 3 · turn 6 · exporter
"...I propose a 'Shared Compliance Reserve' where 1.5% of the FOB
price is held in an escrow account to cover potential EUDR administrative
fines, but this fund must be fully liquidated and paid back to us every
90 days if no infractions occur."

Named instrument · quantified parameter · a release condition. Scorable.
```

That is a term a negotiator can accept, counter or refuse. It has a name, a number and a trigger. It can be diffed against a reservation value. Nothing in Run 1 or Run 2 could be.

SECTION 06

Observed but not sealed

Two further observations from the session are reported here and appear in no chart, because I did not capture them to a file that you could re-check.

- **Response latency.** Wall-clock time per turn fell substantially between Run 1 and Run 3 — roughly 5.4 seconds to roughly 2.4 seconds at the median, n=6 each — and the slowest responses in the baseline correlated with a `factual_grounding_used: true` flag in the payload. The timings were printed to a console and not written to disk, so I present them as an impression with a number attached, not as a measurement.

- **A reservation value disclosed unprompted.** On one early run a simulated counterparty volunteered its own 90-day ceiling in its first reply, before any pressure had been applied. If that reproduces, it is the most commercially serious defect I saw — a counterparty that gives away its limit for free cannot train anyone — and it is precisely the failure the arbiter architecture exists to prevent, by keeping reservation values out of the generation model entirely. I did not seal that response, so I cannot show it to you. **It should be re-tested deliberately before the pilot.**

I am spending a page of a document addressed to a prospective partner on the two things I cannot prove, and that is deliberate. The product's central claim is that a decision rehearsal can be audited. A report about that product which blurred the line between what it measured and what it noticed would be arguing against its own thesis.

SECTION 07

What is genuinely hard here, and what is not

Having worked inside it rather than read about it, my view of where the engineering difficulty actually sits changed.

- **Generating a convincing counterparty is the easy half.** The platform was already doing this well in Run 1 — correct EUDR articles, correct commercial instincts, fluent under pressure. This is table stakes now, and it is not where the defensibility lives.
- **Adjudicating the exchange is the hard half, and it is a different discipline.** Scoring concession discipline requires extracting an offer from free text, binding it to the right variable when several numbers compete in one sentence, and diffing it against a threshold the generating model must never see. That is parsing and arbitration, not language modelling, and it is where the real intellectual property is.
- **The separation of those two halves is the architecture's load-bearing idea.** One record compiles twice: a *phenotype* that goes to the generation layer carrying zero reservation values, and an *arbiter genome* that never leaves the database and carries every threshold. Finding 1 and the unsealed leak in Section 06 are both instances of what goes wrong when that boundary is not enforced end to end.
- **Behavioural parameters belong in physics, not prose.** Friction level is not an instruction to the model to act more impatient. It compiles to interruption and turn-taking parameters. An instruction can be ignored by a language model; a parameter cannot.
- **Video rendering should be bought. Live adjudication has to be built, and it is the hardest open problem in the system.** The avatar, the voice and the lip-sync have several credible vendors and get better every quarter without anyone here. Judging a spoken exchange while it is still happening does not: the time budget falls from seconds to a few hundred milliseconds, offer extraction has to survive speech recognition, and the question of what exactly gets sealed has no settled answer yet. That is where the next real engineering lives.

There is also a lesson in Run 2 that has nothing to do with negotiation and everything to do with how this system is going to be worked on by more than one person. **Behaviour in this product is distributed across a front-end bundle and a set of edge functions, and the deploy surfaces for those two are different.** A change to what the counterparty says can live in either. Until a deploy

makes it unambiguous which prompt is actually being served, every behavioural fix carries the Run 2 risk: it looks shipped, and it is not. If the two organisations end up working on this together, that is the first thing I would want agreed.

SECTION 08

What I would not paper over

OPEN RISK	WHERE IT STANDS
Sample size	Three runs of six turns. Enough to catch a defect and confirm a fix; not a benchmark. Nothing here should be quoted as a performance claim.
Transcript seal	The hashing rule is specified and I have sealed a transcript under it. The platform does not yet reproduce the hash independently. Until it does, the seal is a convention between us rather than a property of the system.
Reservation-value leak	Observed once, not sealed, not reproduced. Untested.
Scoring transparency	Replica adherence is deliberately excluded from the composite score, and unmeasured axes are excluded rather than zero-filled. Both are correct decisions that look like evasions until they are explained. They will need explaining every single time.
Concentration	One engineer holds the whole architecture. That is why this session was possible at the speed it happened, and it is the single largest risk to anyone depending on it.

CLOSING

What I would tell an engineer joining this

The most impressive thing I saw in this session was not the platform. It was the loop. A defect was found by measurement rather than by opinion, a fix was written, the fix was measured, **the measurement said the fix had not landed**, the real cause was found in a different deploy surface entirely, and the next measurement cleared every criterion that had been set in advance. That whole cycle ran inside a single working day, against a live system, with the criteria fixed before the results were known.

Plenty of teams could build a counterparty that argues about the EU Deforestation Regulation. Far fewer would have caught that Run 2 changed nothing, and fewer still would have written down what counted as success before finding out whether they had achieved it.

That discipline is the thing worth partnering with. The code is the easy part to replace.

Prepared for Python Technologies · Najib, Saad and Zaryab. All figures recomputed from sealed artifacts at time of writing. Companion documents: *Banana Pilot Brief*, *Cavendish / EUDR case file v3*, *Integration Brief*.